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Dissecting Sequence, Structure, and Data Recency Effects in Enzyme Commission Prediction under Temporal Distribution Shift

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Abstract

Motivation: Enzyme Commission (EC) prediction is increasingly evaluated with large protein language models and structure-derived features, but headline benchmark scores often conflate model architecture, training-set recency, sequence overlap, label-vocabulary coverage, and temporal distribution shift. This makes it difficult to determine whether a new representation improves biological generalisation or mainly benefits from benchmark design.

Results: We analyze Contact-EC, a gated additive ESM-2/contact-map fusion model, as a controlled probe for decomposing temporal EC prediction performance. On a Swiss-Prot 2023-01 temporal holdout ($N = 124$ complete known-EC proteins), fusion improves Level-4 micro F1 from 0.4508 ± 0.0203 (ESM-2-only) and 0.4244 ± 0.0207 (contact-only) to 0.6241 ± 0.0170 across three seeds. Contact maps alone match ESM-2 650M on sequence-disjoint hard validation (0.7650 vs. 0.7655), and topology perturbations collapse the contact-only model to zero, supporting genuine structural signal. A label-shift audit shows that 344/468 temporal proteins contain partial or novel EC labels outside the evaluated closed-set vocabulary. Data-recency retraining recovers +11.8 pp, whereas end-to-end ESM-2 fine-tuning reduces temporal micro F1 by 5.1 pp. HIT-EC remains stronger in absolute temporal performance (0.8471), indicating that structure-aware fusion is not sufficient by itself to close the temporal gap. A Foldseek/TM-score-disjoint split further shows that absolute novel-fold-like Level-4 performance is low, but validation-calibrated fusion remains higher than ESM-2-only and contact-only baselines across three training seeds (micro F1 0.1685 ± 0.0334 vs. 0.0607 ± 0.0009 and 0.0654 ± 0.0016). The contribution is therefore a reproducible decomposition of when structure helps and when fold shift, benchmark recency, and label shift dominate.

Availability and Implementation: Code and reproducibility artifacts are available at `https://github.com/jamesksh0130/contact-ec`**.**

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Supplementary information: Supplementary data are available at Bioinformatics online.

Key words enzyme commission number; contact map; protein language model; temporal evaluation; sequence–structure fusion; out-of-distribution generalisation

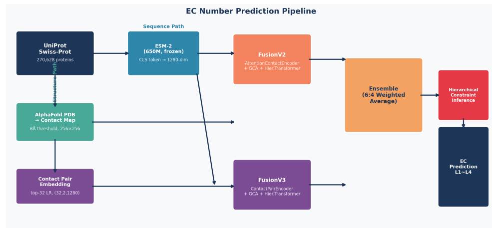

Figure 1 Contact-EC study design for sequence–structure fusion and temporal benchmark decomposition.

1. Introduction

Enzyme Commission (EC) numbers provide a four-level hierarchy for classifying enzyme-catalysed reactions, from broad reaction class to substrate-specific function. Accurate EC assignment supports metabolic reconstruction, pathway annotation, enzyme engineering, and drug-target discovery, but experimental characterisation remains slow relative to the growth of protein sequence databases.

Recent protein language model (PLM) approaches have improved EC prediction substantially. ESM-2 (Lin et al., 2023) encodes residue-level evolutionary and structural information from large sequence corpora, while systems such as HIT-EC (Dumontet et al., 2026) and CLEAN (Yu et al., 2023) use hierarchical or contrastive sequence representations. Structure-aware approaches, including CLEAN-Contact (Ayres et al., 2024), add contact-map features derived from AlphaFold structures (Jumper et al., 2021; Varadi et al., 2022). However, reported scores are often obtained on same-distribution or near-contemporaneous splits. This creates a reliability problem for biological deployment: a model may rank highly because it has access to more recent annotations, an easier closed-set label vocabulary, or near-duplicate sequences, rather than because it generalises to newly annotated enzymes.

We study these factors using Contact-EC, a gated sequence–structure fusion model combining ESM-2 650M embeddings with ResNet-50 contact-map features. The goal is not to claim a new state-of-the-art EC predictor: HIT-EC remains stronger in absolute temporal F1. Instead, Contact-EC is used as an instrument for benchmark decomposition. We ask three questions that are central for Bioinformatics-scale model assessment. First, how different are same-distribution, sequence-disjoint, and temporal EC evaluations? Second, do contact maps contribute signal complementary to ESM-2 after controlling for topology-destroying perturbations? Third, how much of temporal performance is explained by data recency, closed-set label coverage, and fine-tuning strategy rather than by architecture alone?

Our analysis makes six contributions. We provide a temporal label-shift audit for Swiss-Prot 2023-01, separating 124 complete known-EC proteins from 344 partial or novel-label cases. We show that contact maps alone match ESM-2 on hard sequence-disjoint validation, while fusion improves temporal micro F1 over either single modality. We verify that the contact branch uses fold topology rather than contact-map density through shuffled, diagonal-only, and density-matched controls. We add a Foldseek/TM-score-disjoint split to test structural generalisation beyond sequence-disjoint evaluation. We decompose data recency and fine-tuning, finding that newer Swiss-Prot data helps whereas end-to-end fine-tuning over-adapts. Finally, we translate these results into reporting requirements for EC prediction studies: temporal cutoff, sequence overlap, fold-level relatedness, label-vocabulary coverage, partial-label frequency, structure availability, and whether each result is measured on a closed-set or label-shifted benchmark.

2. System and Methods

2.1. Data and Splits

The primary controlled experiments follow EC-Bench: Swiss-Prot February 2018 for training (201,865 proteins after local preprocessing) and Swiss-Prot 2023-01 for temporal testing. Sequences were truncated to 1,024 residues. AlphaFold structures were retrieved from the AlphaFold Protein Structure Database (Varadi et al., 2022) when available; proteins without structures received zero contact maps. Contact maps were built from Cα distances using an 8 Åthreshold, resized to 256×256, and encoded in three channels: all contacts, short-range contacts ($|i - j| < 12$), and long-range contacts ($|i - j| \geq 12$).

We distinguish four evaluation regimes. A random 80/10/10 Swiss-Prot split measures in-distribution performance. A MMSeqs2 (Steinegger and Söding, 2017) 30% identity cluster split measures sequence-disjoint generalisation. An EC-Bench-style hard validation set (`val_hard`; 6,247 proteins, ≤30% sequence identity) provides a shared validation protocol rather than an independent hidden test. Of 468 Swiss-Prot 2023-01 temporal proteins, only 124 have complete Level-4 EC labels inside the SP-2018 vocabulary; the remaining 344 contain partial or novel labels. A final temporal audit found no accession or exact-sequence overlap between these 124 proteins and either SP-2018 or ExpA training, but temporal evaluation alone does not guarantee fold-disjointness.

To directly evaluate fold-level shift, we constructed an additional Foldseek-disjoint split from the SP-2018 training corpus. Proteins with available AlphaFold structures were clustered by Foldseek (van Kempen et al., 2024) using TM-score threshold 0.50, coverage 0.80, alignment type 1, cluster mode 1, and sensitivity 7.5. Cluster assignment produced 170,573 training, 25,565 validation, and 16,725 test proteins across 3,051/381/381 structure clusters, with zero protein-ID overlap across partitions. This split is used only for the fold-disjoint audit and all models are retrained on the corresponding training partition.

2.2. Model

The sequence branch uses ESM-2 650M (Lin et al., 2023) (`facebook/esm2_t33_650M_UR50D`) and extracts the 1280-dimensional `[CLS]` embedding. The structure branch applies a ResNet-50 encoder (He et al., 2016) with attention pooling to the three-channel contact map and projects the output to 512 dimensions. The two representations are projected to a shared 1024-dimensional space and combined by gated sequence–structure fusion:

$$\tilde{\mathbf{e}} = \phi(W_e \mathbf{e}), \qquad \tilde{\mathbf{s}} = W_s \mathbf{s},$$

$$\mathbf{g} = \sigma(W_g[\tilde{\mathbf{e}}; \tilde{\mathbf{s}}]), \qquad \mathbf{f} = \tilde{\mathbf{e}} + \mathbf{g} \odot W_a \tilde{\mathbf{s}}.$$

This gated additive fusion modulates a projected structure vector by sequence context and allows fallback when the contact map is missing or uninformative. Contact-EC-Hier passes the fused vector through a two-layer Transformer over four EC-level tokens; the flat variant uses fully connected sigmoid outputs. To audit whether this learned fusion block is necessary, we also train three same-encoder flat-head controls: concatenation followed by an MLP, element-wise summation, and a feature-wise gated MLP (Supplementary architecture-baseline table).

2.3. Training and Evaluation

Models are trained with weighted binary cross-entropy across EC levels, with weights $(0.1, 0.1, 0.2, 0.6)$ for Levels 1–4 and masks for partial annotations. Phase 1 trains downstream modules for 30 epochs with frozen ESM-2; Phase 2 partially unfreezes the final two ESM-2 blocks for 20 epochs at a lower learning rate. AdamW (Loshchilov and Hutter, 2019) with cosine annealing is used throughout, and checkpoints are selected only on validation data. Level-4 micro F1 is the primary metric, with macro F1, weighted F1, rare-EC F1, precision, and recall reported for interpretation. Hyper-parameters were fixed before temporal testing.

3. Experiments and Results

3.1. EC-Bench-Style Validation

On EC-Bench `val_hard`, the checkpoint-selected hard validation protocol, contact maps alone reach nearly identical Level-4 micro F1 to ESM-2 650M (0.7650 vs. 0.7655; Table 1). This parity suggests that fold topology captures substantial EC-discriminative information even without sequence embeddings. Fusion improves substantially: Contact-EC reaches 0.8879, while Contact-EC-Hier reaches 0.8698. Training dynamics in Supplementary Fig. S1 show that B3 converges within about 10 epochs and that fusion continues to improve during partial unfreezing.

Table 1 EC-Bench hard validation results (`val_hard`, 6,247 proteins, $\leq$30% identity), used for validation/checkpoint selection.

Model	L4 Micro F1	L4 Macro F1	Best Epoch
B1 (ESM-2, flat FC)	0.7655	0.1353	30
B2 (ESM-2, hier. FC)	0.4204	0.0383	30
B3 (Contact map only)	0.7650	0.1335	10
Contact-EC (flat FC)	**0.8879**	**0.2294**	29
Contact-EC-Hier	0.8698	0.2141	20 (Phase 2)

3.2. Temporal Label Shift

The official Swiss-Prot 2023-01 temporal set contains 468 proteins, but only 124 are complete known-EC cases that can be scored within the SP-2018 closed-set label space (Table 2). The remaining 344 proteins expose an important benchmark limitation: Level-4 closed-set scoring is undefined for incomplete partial EC labels and cannot assign novel Level-4 classes outside the training vocabulary. We therefore report main temporal comparisons on the 124 evaluable proteins and use the full 468 proteins as a label-shift audit. This separation is essential for interpreting temporal EC prediction: a low score on all 468 proteins may reflect open-vocabulary label shift rather than a failure to rank known EC classes, whereas a high score on the 124-protein subset may overstate deployability on newly annotated enzymes. Supplementary hierarchical-prefix evaluation further shows that partial annotations retain useful lower-level supervision and should not be counted as Level-4 failures.

Table 2 Swiss-Prot 2023-01 label-shift audit. Partial annotations lack complete Level-4 ground truth and are not assigned Level-4 F1.

Subset	N	Interpretation	Closed-set L4 status
Complete known EC labels	124	Main evaluable subset	Scored: 0.6032
Partial EC annotations	309	Incomplete Level-4 labels	Not scored at L4
Novel EC labels	35	Outside SP-2018 vocabulary	Out of vocabulary
All temporal proteins	468	Label-shift audit	Coverage audit

3.3. Temporal Performance

On the complete temporal subset, sequence–structure fusion outperforms both single-modality baselines (Table 3). Across three seeds, Contact-EC improves micro F1 from 0.4508$\pm$0.0203 for B1 and 0.4244$\pm$0.0207 for B3 to 0.6241$\pm$0.0170. The 3B flat-fusion variant reaches 0.6316 in a single run, a modest gain over the 650M fusion mean relative to the 4.5$\times$ parameter increase. Contact-EC-Hier improves over single-modality baselines but underperforms the flat head, mainly through lower recall. HIT-EC remains higher at 0.8471, so we interpret Contact-EC as a decomposition model rather than a replacement for stronger sequence-only systems.

Case-wise and baseline audits support this framing. Among 124 temporal proteins, HIT-EC and Contact-EC both recover at least one correct EC for 63 proteins, HIT-EC alone recovers 45, Contact-EC alone recovers 2, and both miss 14. Changing the global threshold only modestly changes Contact-EC micro F1 (0.6032 at 0.5 versus 0.6167 at 0.65), and MMseqs2 top-hit EC transfer reaches 0.5852. Trainable fusion controls remain below Contact-EC: concatenation reaches 0.5922±0.0150 micro F1, summation 0.5505±0.0230, and a gated MLP 0.5543±0.0525 (Supplementary architecture-baseline table).

Table 3 Temporal and external benchmark results. Swiss-Prot 2023-01 uses the complete known-EC subset ($N = 124$). B1, B3, and Contact-EC flat FC temporal values are mean ± sample standard deviation across three seeds; other rows are single-run or externally reported. Bold denotes the best value among our SP-2018-trained models.

	Swiss-Prot 2023-01			Price-149
Model	Micro F1	Weighted F1	Rare-EC F1	Micro F1
HIT-EC (own split)	0.930	–	–	–
HIT-EC (this test)	0.8471	0.8134	0.8322	–
CLEAN-Contact	–	–	–	0.525
B1 (ESM-2, flat FC)	0.4508 ± 0.0203	0.3387 ± 0.0194	0.3292 ± 0.0272	0.0324
B2 (ESM-2, hier. FC)	0.1111	0.0835	0.0000	0.0000
B3 (Contact map only)	0.4244 ± 0.0207	0.3325 ± 0.0237	0.3354 ± 0.0254	0.0000
Contact-EC (flat FC)	0.6241 ± 0.0170	0.5278 ± 0.0222	$\mathbf{0.5257 \pm 0.0313}$	0.2500
Contact-EC-Hier	0.5690	0.4528	0.3621	0.1263
Contact-EC-3B (flat FC)	**0.6316**	**0.5477**	0.4786	**0.2915**
Contact-EC-ExpA (recent data)	0.7209	0.6578	0.5283	0.2764
Contact-EC-E2E (fine-tuned)	0.6703	0.6382	0.5091	0.2522

Bootstrap confidence intervals (1,000 resamples) confirm the main differences: Contact-EC-3B [0.547, 0.711], Contact-EC [0.511, 0.686], Contact-EC-Hier [0.476, 0.664], B1 [0.315, 0.521], and B3 [0.254, 0.473]. McNemar tests further support Contact-EC over B1 ($\chi^2 = 22.4$, $p < 0.001$) and over B3 ($\chi^2 = 37.0$, $p < 0.001$). Supplementary late-fusion diagnostics show that simple post-hoc averaging or maximum pooling of B1 and B3 probabilities does not reproduce the learned fusion gain (best late-fusion temporal micro F1 = 0.4673 vs. 0.6032 for Contact-EC). Retrained simple fusion baselines also remain lower than Contact-EC (Supplementary architecture-baseline table).

3.4. Foldseek-Disjoint Generalisation

Sequence-disjoint evaluation does not necessarily remove structural fold similarity. On the primary Foldseek/TM-score 0.50 split, all models show a large absolute drop. Across three training seeds, the default Level-4 threshold of 0.5 gives micro F1 of 0.0433 ± 0.0042 for B1, 0.0400 ± 0.0054 for B3, and 0.0872 ± 0.0122 for fusion. Validation-selected thresholds applied once to the held-out Foldseek test split preserve the same ordering, with fusion reaching 0.1685 ± 0.0334 micro F1 versus 0.0607 ± 0.0009 for B1 and 0.0654 ± 0.0016 for B3. A single-seed TM-score sweep at 0.40/0.50/0.60 also preserves the fusion advantage (0.2981/0.0998/0.4403 micro F1), although the 0.50 split has lower label coverage and a different cluster-assignment mode (Supplementary Tables S13–S15). In contrast, MMseqs2 top-hit EC transfer is much stronger on the same split (Level-4 micro F1 = 0.5393), showing that this audit tests complementary neural generalisation rather than replacement of curated homology transfer.

The split also induces label-coverage shift: 31.5% of positive Level-4 labels in Foldseek test are absent from the corresponding training partition, and 31.6% of test proteins contain at least one unseen Level-4 label. Among seen labels, fusion is strongest for moderately represented classes (0.2484 micro F1 for 6–25 training proteins and 0.3278 for 26–100), indicating that fold-disjoint performance reflects both structural shift and label-frequency shift. A three-seed failure-mode audit further shows that fusion rescues 17.3% of Foldseek-test proteins missed by B1 and 12.5% missed by B3; 11.0% are fusion-only rescues, whereas 41.9% of all-model failures contain an unseen true Level-4 label.

3.5. Topology Controls and Data Recency

To test whether the contact branch uses topology rather than contact-map density or length proxies, we evaluated the B3 checkpoint under shuffled, diagonal-only, density-matched random, and all-zero contact maps. B3 collapses to micro F1=0.000 under all perturbations, whereas Contact-EC-Hier remains robust by falling back to the ESM-2 branch (Table 4). Thus contact-only performance depends on structured topology, not superficial map statistics.

Table 4 Contact-map perturbation controls on the random-split test set.

Contact map variant	B3 Micro F1	Contact-EC-Hier Micro F1
Original contact maps	0.8987	0.9261
Shuffled contacts	0.0000	0.9176
Diagonal-only contacts	0.0000	0.9138
Density-matched random	0.0000	0.9152
All-zero maps	0.0000	0.9124

The recent-corpus experiment tests the effect of updating the training corpus and label vocabulary. Contact-EC-ExpA improves from 0.6032 to 0.7209, but sample size, class frequencies, and homolog coverage also change, so this is a recent-corpus effect rather than a pure date effect. In contrast, end-to-end ESM-2 fine-tuning reaches similar in-distribution validation performance but drops to 0.6703 temporally. This suggests over-adaptation of the PLM backbone under limited supervised data and supports frozen or lightly tuned representations for temporal robustness. The +11.8 pp recent-corpus gain is larger than the +2.8 pp gain from replacing ESM-2 650M with the 3B backbone, reinforcing that benchmark date, data volume, homolog coverage, and label coverage can matter more than backbone scale.

4. Discussion

The results support a conservative interpretation. Contact maps provide a genuine and complementary signal for EC prediction, especially on sequence-disjoint validation, moderately represented fold-disjoint labels, and several broad EC families, but explicit structure does not by itself close the gap to strong sequence-only systems trained on newer data. The strongest temporal score in this study remains HIT-EC, and controlled retraining shows that recent-corpus factors explain a substantial part of the remaining gap. This distinction matters for method development: an architecture that improves a fixed-date, closed-set benchmark may still underperform a simpler or sequence-only system trained with a more recent label vocabulary.

The evaluation also shows why EC benchmark design matters. The same model can look strong on random splits, near-parity on sequence-disjoint validation, and much weaker under temporal shift. Moreover, the full Swiss-Prot 2023-01 temporal set is not a single closed-set benchmark: 344/468 proteins contain partial or novel labels. Future studies should report training cutoff, sequence overlap, fold-level relatedness, label-vocabulary coverage, partial-label frequency, and structure availability alongside headline F1. We recommend reporting both the evaluable closed-set subset and the full temporal label-shift set, because these answer different questions: recognition of known EC classes versus robustness to newly curated or incomplete annotations.

There are limitations. Contact-EC relies on AlphaFold or predicted structures, so performance can degrade when external structures or predicted contact maps come from a shifted distribution. On Price-149, supplementary audit results show that input coverage is available but exact Level-4 specificity collapses despite stronger coarse-level EC recovery, indicating an OOD calibration and specificity failure rather than a simple missing-input problem. A calibration audit supports this interpretation: on Price-149, Contact-EC emits nonempty predictions for 51.5% of encoded proteins with median top-1 probability 0.5527, but top-1 hit rate is only 27.9%. Supplementary contact-pair and raw contact-map topology nearest-neighbor audits further show high nearest training-set similarity for nearly all evaluable temporal proteins, explaining why temporal results should not be interpreted as fold-disjoint structural generalization. The explicit Foldseek split addresses this issue, but it also exposes severe label-coverage shift: in the primary 0.50 split, 31.5% of Foldseek-test positive labels are unseen in the corresponding training partition, compared with 15.9% and 10.3% in the 0.40 and 0.60 sweep splits. MMseqs2 top-hit transfer remains substantially stronger than all closed-set neural models, reinforcing that Contact-EC is a decomposition probe rather than a homology replacement. Macro F1 remains low due to the long-tailed Level-4 label space. The temporal test is small; although the main temporal B1/B3/Contact-EC comparison and the Foldseek audit now include three-seed repeats, 3B, recent-data, and end-to-end variants are still single-run analyses due to compute cost. Bootstrap and paired tests therefore quantify sample-level uncertainty rather than definitive rankings. Finally, supplementary open-vocabulary audits show that fusion confidence can partially flag unseen-label cases, but closed-set models cannot solve novel EC discovery without retrieval, abstention, or label-expansion mechanisms.

5. Conclusion

This study reframes structure-aware EC prediction as a benchmark decomposition problem. Contact maps alone can match ESM-2 650M on hard sequence-disjoint validation, and sequence–structure fusion substantially improves temporal F1 over either modality alone. Perturbation controls confirm that the structure branch uses fold topology, and Foldseek-disjoint evaluation shows that fusion retains a measurable advantage under fold-level shift even though absolute closed-set Level-4 performance remains low. Data-recency and fine-tuning controls show that annotation cutoff and backbone adaptation strongly affect temporal performance. These findings argue for EC prediction benchmarks that explicitly separate representation generalisation from fold shift, label shift, data recency, backbone scale, and structure availability.

Data Availability

Swiss-Prot sequences and EC annotations were downloaded from UniProt (`https://www.uniprot.org`). AlphaFold structures were retrieved from the AlphaFold Protein Structure Database (`https://alphafold.ebi.ac.uk`). New-392 and Price-149 benchmark files are available from the CLEAN-Contact repository (Ayres et al., 2024). EC-Bench training and test data can be obtained from the EC-Bench GitHub repository (Davoudi et al., 2026). Code and reproducibility artifacts for reproducing tables and figures are available at `https://github.com/jamesksh0130/contact-ec`.

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Conflict of Interest

The author declares no competing interests.

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Author Contributions

CRediT author statement: Seunghyon Kim: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Visualization, Writing – Original Draft, Writing – Review & Editing. The author has read and approved the final manuscript.

References

Lindsey B. Ayres et al. CLEAN-Contact: Contrastive learning-enabled enzyme functional annotation prediction with structural inference. *Communications Biology*, 7:1620, 2024. doi: 10.1038/s42003-024-07359-z.

Saeedeh Davoudi, Christopher S. Henry, Christopher S. Miller, and Farnoush Banaei-Kashani. EC-Bench: a benchmark for enzyme commission number prediction. *Bioinformatics Advances*, 6(1): vbag004, 2026. doi: 10.1093/bioadv/vbag004.

Louis Dumontet, So-Ra Han, Jun Hyuck Lee, Tae-Jin Oh, and Mingon Kang. Trustworthy prediction of enzyme commission numbers using a hierarchical interpretable transformer. *Nature Communications*, 17:1146, 2026. doi: 10.1038/s41467-026-68727-3.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016. doi: 10.1109/CVPR.2016.90.

John Jumper et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596: 583–589, 2021. doi: 10.1038/s41586-021-03819-2.

Zeming Lin et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. URL `https://openreview.net/forum?id=Bkg6RiCqY7`.

Martin Steinegger and Johannes Söding. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology*, 35:1026–1028, 2017. doi: 10.1038/nbt.3988.

Michel van Kempen, Stephanie S. Kim, Christoph Tumescheit, Milot Mirdita, Jeongjae Lee, Cameron L. M. Gilchrist, Johannes Söding, and Martin Steinegger. Fast and accurate protein structure search with Foldseek. *Nature Biotechnology*, 42:243–246, 2024. doi: 10.1038/s41587-023-01773-0.

Mihaly Varadi et al. AlphaFold protein structure database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Research*, 50(D1): D439–D444, 2022. doi: 10.1093/nar/gkab1061.

Tianhao Yu et al. Enzyme function prediction using contrastive learning. *Science*, 379(6639): 1358–1363, 2023. doi: 10.1126/science.adf2465.